P2 Chapter 10: Numerical Methods

Modelling Applications

Application to Modelling

[Textbook] The price of a car in £s, x years after purchase, is modelled by the function

$$f(x) = 15\,000\,(0.85)^x - 1000 \sin x\,, \qquad x > 0$$

(a) Use the model to find the value, to the nearest hundred £s, of the car 10 years after purchase.

(b) Show that $f(x)$ has a root between 19 and 20.

(c) Find $f'(x)$

(d) Taking 19.5 as a first approximation, apply the Newton-Raphson method once to $f(x)$ to obtain a second approximation for the time when the value of the car is zero. Give your answer to 3 decimal places.

(e) Criticise this model with respect to the value of the car as it gets older.

a

?

b

?

c

?

d

?

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(e) Criticise this model with respect to the value of the car as it gets older.

a $f(10) = 3497.13$
$\rightarrow$ £3500

b $f(19) = 543.11 > 0$
$f(20) = -331.55 < 0$
There is a change of sign between 19 and 20, so that is a root in the interval $19 < x < 20$

c $f'(x) = (15\,000)(0.85)^x(\ln 0.85) - 1000\cos x$
$f(19.5) = 25.0693\ldots$
$f'(19.5) = -898.3009\ldots$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 19.528$$

d In reality, the car can never have a negative value so this model is not reasonable for cars that are approximately 20 or more years old.

Exercise 10.4

Pearson Pure Mathematics Year 2/AS
Page 86

Homework Exercise

1 An astronomer is studying the motion of a planet moving along an elliptical orbit. She formulates the following model relating the angle moved at a given time, E radians, to the angle the planet would have moved if it had been travelling on a circular path, M radians:

$$M = E - 0.1 \sin E,\ E \geqslant 0$$

In order to predict the position of the planet at a particular time, the astronomer needs to find the value of E when $M = \frac{\pi}{6}$

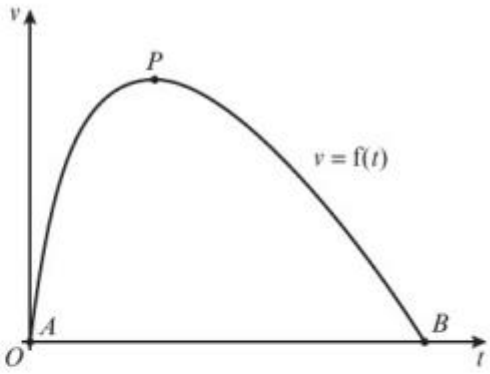
a Show that this value of E is a root of the function $f(x) = x - 0.1 \sin x - k$ where k is a constant to be determined.

b Taking 0.6 as a first approximation, apply the Newton–Raphson procedure once to $f(x)$ to obtain a second approximation for the value of E when $M = \frac{\pi}{6}$

c By considering a change of sign on a suitable interval of $f(x)$, show that your answer to part **b** is correct to 3 decimal places.

Homework Exercise

2 The diagram shows a sketch of part of the curve with equation $v = \mathrm{f}(t)$, where $\mathrm{f}(t) = \left(10 - \frac{1}{2}(t+1)\right)\ln(t+1)$. The function models the velocity in m/s of a skier travelling in a straight line.



a Find the coordinates of A and B.

b Find $\mathrm{f}'(t)$.

c Given that P is a stationary point on the curve, show that the t-coordinate of P lies between 5.8 and 5.9.

d Show that the t-coordinate of P is the solution to

$$t = \frac{20}{1 + \ln(t+1)} - 1$$

An approximation for the t-coordinate of P is found using the iterative formula

$$t_{n+1} = \frac{20}{1 + \ln(t_n + 1)} - 1$$

e Let $t_0 = 5$. Find the values of t_1, t_2 and t_3. Give your answers to 3 decimal places.

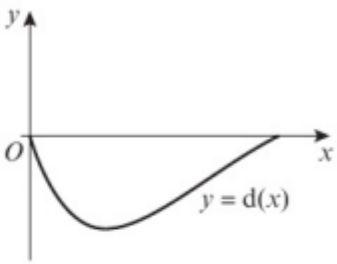
Homework Exercise

3 The depth of a stream is modelled by the function

$$d(x) = e^{-0.6x}(x^2 - 3x),\ 0 \leqslant x \leqslant 3$$

where x is the distance in metres from the left bank of the stream and $d(x)$ is the depth of the stream in metres.

The diagram shows a sketch of $y = d(x)$.



a Explain the condition $0 \leqslant x \leqslant 3$.

b Show that $d'(x) = -\frac{1}{5}e^{-0.6x}(ax^2 + bx + c)$, where a, b and c are constants to be found.

c Show that $d'(x) = 0$ can be written in the following ways:

i $x = \sqrt{\dfrac{19x - 15}{3}}$ **ii** $x = \dfrac{3x^2 + 15}{19}$ **iii** $x = \dfrac{19x - 15}{3x}$

d Let $x_0 = 1$. Show that only one of the three iterations converges to a stationary point of $y = d(x)$, and find the x-coordinate at this point correct to 3 decimal places.

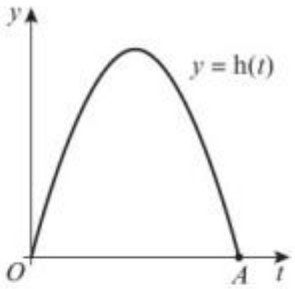
e Find the maximum depth of the river in metres to 2 decimal places.

Homework Exercise

4 Ed throws a ball for his dog. The vertical height of the ball is modelled by the function

$$h(t) = 40\sin\left(\frac{t}{10}\right) - 9\cos\left(\frac{t}{10}\right) - 0.5t^2 + 9,\ t \geqslant 0$$

$y = h(t)$ is shown in the diagram.



a Show that the t-coordinate of A is the solution to

$$t = \sqrt{18 + 80\sin\left(\frac{t}{10}\right) - 18\cos\left(\frac{t}{10}\right)}$$ **(3 marks)**

To find an approximation for the t-coordinate of A, the iterative formula

$$t_{n+1} = \sqrt{18 + 80\sin\left(\frac{t_n}{10}\right) - 18\cos\left(\frac{t_n}{10}\right)}$$ is used.

b Let $t_0 = 8$. Find the values of t_1, t_2, t_3 and t_4. Give your answers to 3 decimal places. **(3 marks)**

c Find $h'(t)$. **(2 marks)**

d Taking 8 as a first approximation, apply the Newton–Raphson method once to $h(t)$ to obtain a second approximation for the time when the height of the ball is zero. Give your answer to 3 decimal places. **(3 marks)**

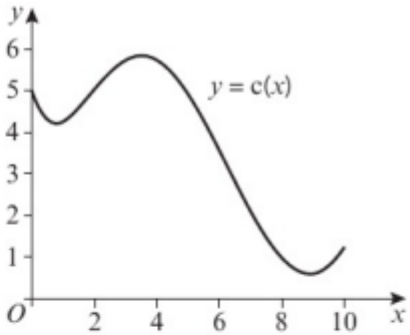
e Hence suggest an improvement to the range of validity of the model. **(2 marks)**

Homework Exercise

5 The annual number of non-violent crimes, in thousands, in a large town x years after the year 2000 is modelled by the function

$$c(x) = 5e^{-x} + 4\sin\left(\frac{x}{2}\right) + \frac{x}{2},\ 0 \leqslant x \leqslant 10$$

The diagram shows the graph of $y = c(x)$.



a Find $c'(x)$. **(2 marks)**

b Show that the roots of the following equations correspond to the turning points on the graph of $y = c(x)$.

i $x = 2\arccos\left(\frac{5}{2}e^{-x} - \frac{1}{4}\right)$ **(2 marks)**

ii $x = \ln\left(\frac{10}{4\cos\left(\frac{x}{2}\right) + 1}\right)$ **(2 marks)**

c Let $x_0 = 3$ and $x_{n+1} = 2\arccos\left(\frac{5}{2}e^{-x_n} - \frac{1}{4}\right)$. Find the values of x_1, x_2, x_3 and x_4. Give your answers to 3 decimal places. **(3 marks)**

d Let $x_0 = 1$ and $x_{n+1} = \ln\left(\frac{10}{4\cos\left(\frac{x_n}{2}\right) + 1}\right)$. Find the values of x_1, x_2, x_3 and x_4. Give your answers to 3 decimal places. **(3 marks)**

A councillor states that the number of non-violent crimes in the town was increasing between October 2000 and June 2003.

e State, with reasons whether the model supports this claim. **(2 marks)**

Homework Answers

1 **a** $\frac{\pi}{6} = E - 0.1\sin E$, if E is a root then $f(E) = 0$

$E - 0.1\sin E - k = 0 \Rightarrow E - 0.1\sin E = k \Rightarrow \frac{\pi}{6} = k$

b 0.5782...

c $f(0.5775) = -0.00069... < 0$, $f(0.5785) = 0.00022 > 0$. Change of sign implies root in interval [0.5775, 0.5785], so root is 0.578 to 3 d.p.

2 **a** $A(0, 0)$ and $B(19, 0)$

b $f'(t) = \frac{10}{t+1} - \left(\frac{\ln(t+1)}{2} + \frac{1}{2}\right)$

c $f'(5.8) = \frac{10}{5.8+1} - \left(\frac{\ln(5.8+1)}{2} + \frac{1}{2}\right) = 0.0121... > 0$

$f'(5.9) = \frac{10}{5.9+1} - \left(\frac{\ln(5.9+1)}{2} + \frac{1}{2}\right) = -0.0164... < 0$

The sign change implies that the speed changes from increasing to decreasing, so the greatest speed of the skier lies between 5.8 and 5.9.

d $f'(t) = \frac{10}{t+1} - \left(\frac{\ln(t+1)}{2} + \frac{1}{2}\right) = 0$

$\frac{\ln(t+1)+1}{2} = \frac{10}{t+1}$

$(t+1)(\ln(t+1)+1) = 20$

$t + 1 = \frac{20}{\ln(t+1)+1}$

$t = \frac{20}{\ln(t+1)+1} - 1$

e $t_1 = 6.164$, $t_2 = 5.736$, $t_3 = 5.879$

3 **a** $d(x) = 0 \Rightarrow x^2 - 3x = 0$

$x(x-3) = 0 \Rightarrow x = 0, x = 3$

The river bed is 3 m wide so the function is only valid for $0 \leqslant x \leqslant 3$.

b $d'(x) = 2xe^{-0.6x} - \frac{3}{5}x^2e^{-0.6x} - 3e^{-0.6x} + \frac{9}{5}xe^{-0.6x}$

$d'(x) = e^{-0.6x}\left(-\frac{3}{5}x^2 + \frac{19}{5}x - \frac{15}{5}\right)$

$d'(x) = -\frac{1}{5}e^{-0.6x}(3x^2 - 19x + 15)$

So $a = 3$, $b = -19$ and $c = 15$

c $-\frac{1}{5}e^{-0.6x} \neq 0$ so $d'(x) = 0 \Rightarrow 3x^2 - 19x + 15 = 0$

i $3x^2 - 19x + 15 = 0 \Rightarrow 3x^2 = 19x - 15$

$\Rightarrow x^2 = \frac{19x - 15}{3} \Rightarrow x = \sqrt{\frac{19x - 15}{3}}$

ii $3x^2 - 19x + 15 = 0 \Rightarrow 3x^2 + 15 = 19x$

$\Rightarrow x = \frac{3x^2 + 15}{19}$

iii $3x^2 - 19x + 15 = 0 \Rightarrow 3x^2 = 19x - 15$

$\Rightarrow x = \frac{19x - 15}{3x}$

d Part **i** and **iii** tend to 5.408... which is outside the required range. Part **ii** tends to $x = 0.924$.

e 1.10 m.

Homework Answers

4 **a** $h(t) = 0$

$$40\sin\left(\frac{t}{10}\right) - 9\cos\left(\frac{t}{10}\right) - 0.5t^2 + 9 = 0$$

$$40\sin\left(\frac{t}{10}\right) - 9\cos\left(\frac{t}{10}\right) + 9 = 0.5t^2$$

$$80\sin\left(\frac{t}{10}\right) - 18\cos\left(\frac{t}{10}\right) + 18 = t^2$$

$$\Rightarrow t = \sqrt{18 + 80\sin\left(\frac{t}{10}\right) - 18\cos\left(\frac{t}{10}\right)}$$

b $t_1 = 7.928,\ t_2 = 7.896,\ t_3 = 7.882,\ t_4 = 7.876$

c $h'(t) = 4\cos\left(\frac{t}{10}\right) + 0.9\sin\left(\frac{t}{10}\right) - t$

d 7.874 (3 d.p.)

e Restrict the range of validity to $0 \leqslant t \leqslant A$

5 **a** $c'(x) = -5e^{-x} + 2\cos\left(\frac{x}{2}\right) + \frac{1}{2}$

b **i** $-5e^{-x} + 2\cos\left(\frac{x}{2}\right) + \frac{1}{2} = 0$

$$\Rightarrow \cos\left(\frac{x}{2}\right) = \frac{5}{2}e^{-x} - \frac{1}{4} \Rightarrow x = 2\arccos\left[\frac{5}{2}e^{-x} - \frac{1}{4}\right]$$

ii $-5e^{-x} + 2\cos\left(\frac{x}{2}\right) + \frac{1}{2} = 0 \Rightarrow e^{-x} = \dfrac{4\cos\left(\frac{x}{2}\right) + 1}{10}$

$$\Rightarrow e^{x} = \frac{10}{4\cos\left(\frac{x}{2}\right) + 1} \Rightarrow x = \ln\left(\frac{10}{4\cos\left(\frac{x}{2}\right) + 1}\right)$$

c $x_1 = 3.393,\ x_2 = 3.475,\ x_3 = 3.489,\ x_4 = 3.491$

d $x_1 = 0.796, x_2 = 0.758, x_3 = 0.752, x_4 = 0.751$

e The model does support the assumption that the crime rate was increasing. The model shows that there is a minimum point $\frac{3}{4}$ of the way through 2000 and a maximum point mid-way through 2003. So, the crime rate is increasing in the interval between October 2000 and June 2003.